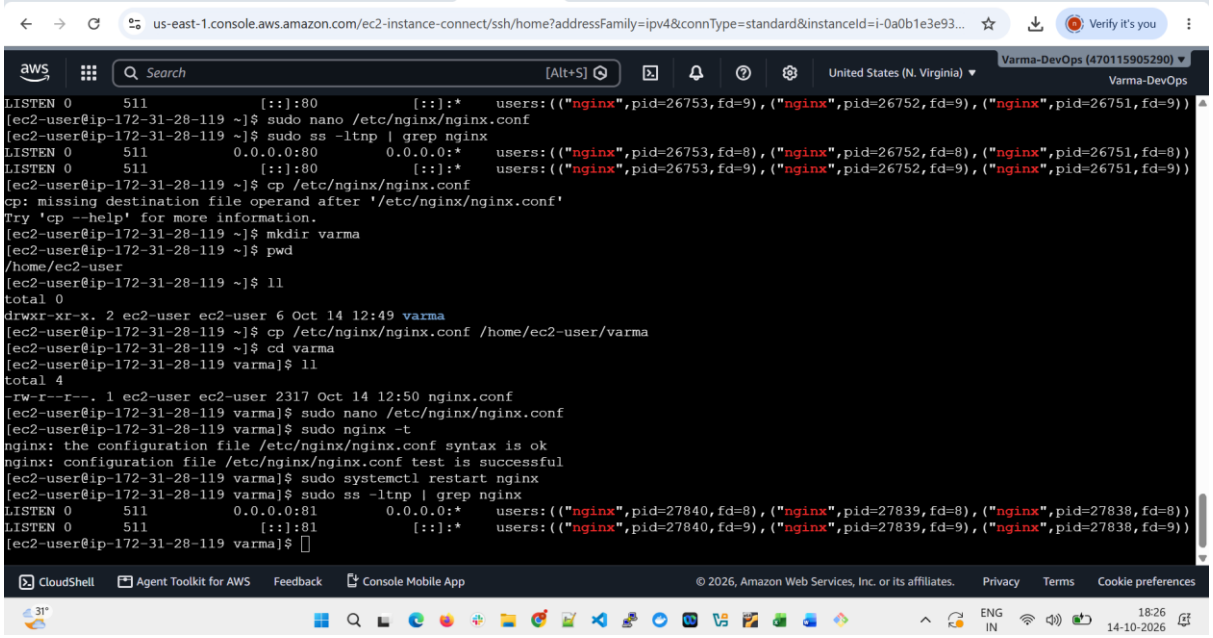
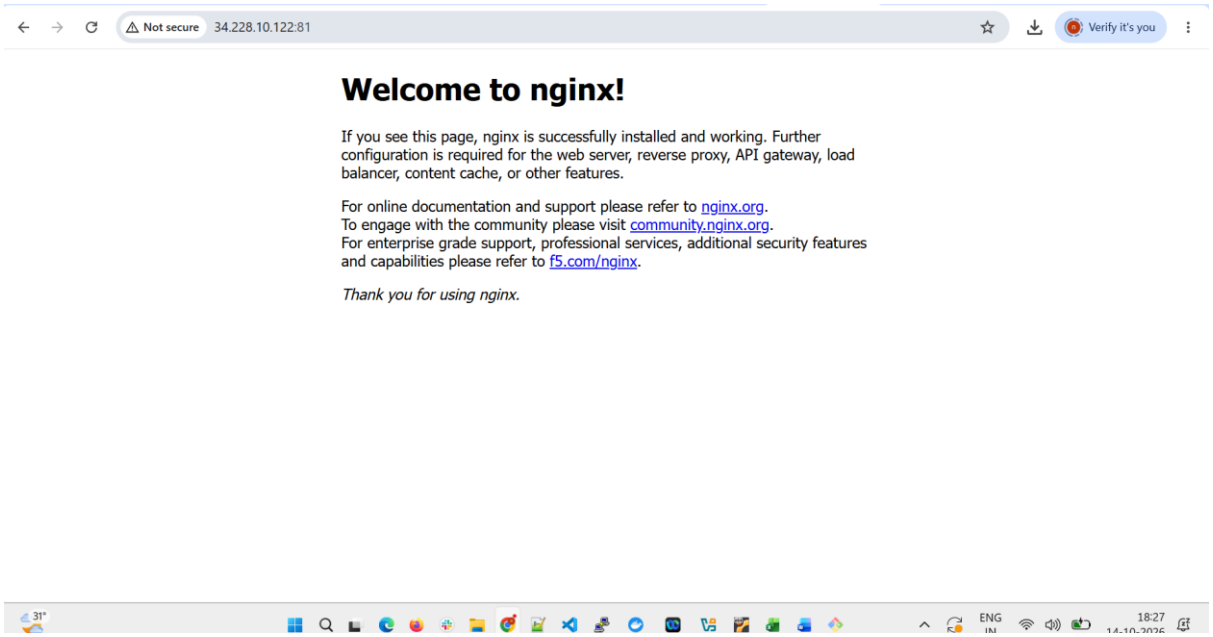
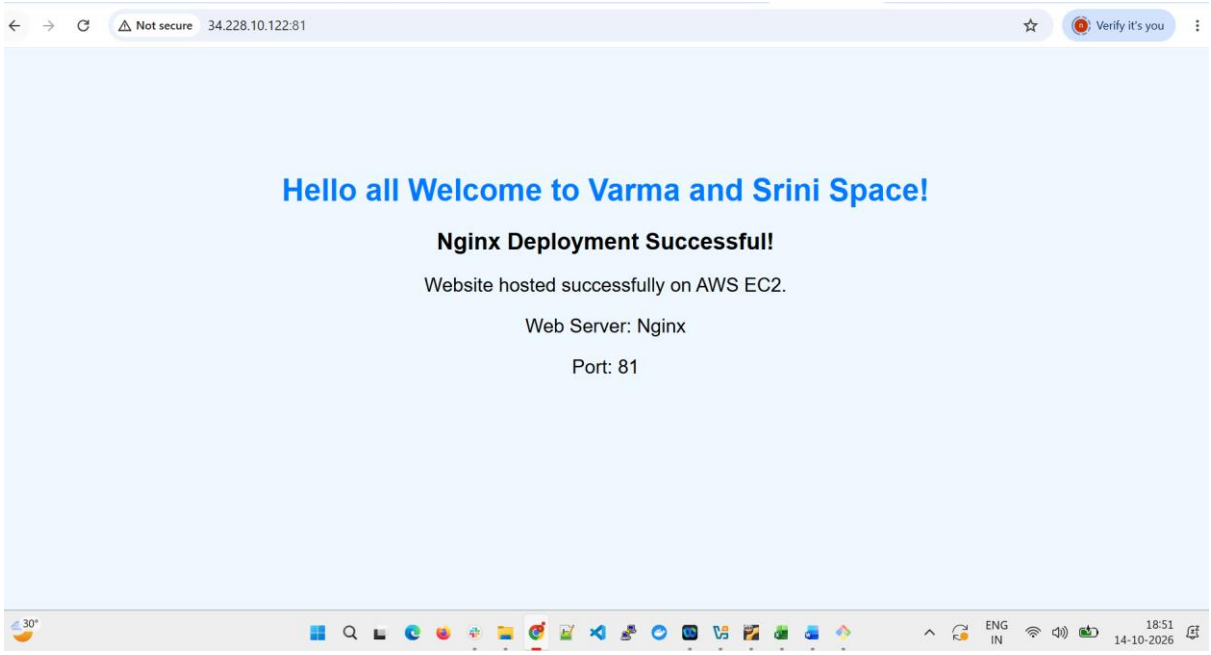
1. Install nginx and run nginx on port number 81.





2. Deploy a sample index.html file on nginx.



3. Install Apache and run Apache on port number 82

```
< Content-Length: 0
< Content-Type: text/html
<
* Connection #0 to host 127.0.0.1:82 left intact
root@VM1:~# hostname -i
fd17:625c:f037:2:a00:27ff:fe2d:d9a3 172.17.0.1 10.0.2.15 fe80::a00:27ff:fe2d:d9a3
root@VM1:~# hostname -i
fd17:625c:f037:2:a00:27ff:fe2d:d9a3 172.17.0.1 192.168.56.101 fe80::a00:27ff:fe2d:d9a3
root@VM1:~# sudo ufw status
Status: inactive
root@VM1:~# sudo ss -ltnp 'sport = :82'
State              Recv-Q             Send-Q                         Local Address:Port                          Peer Address:Port
Process
LISTEN             0                  511                                  *:82                                       *:*
 users:(("apache2",pid=9538,fd=4),("apache2",pid=9537,fd=4),("apache2",pid=9534,fd=4))
root@VM1:~# ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
3: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
       valid_lft 408sec preferred_lft 408sec
root@VM1:~# echo '<h1> Welcome to my worl Rishi</h1><p>This is running in port 82!</p>' | sudo tee /var/www/html/index.html
<h1> Welcome to my worl Rishi</h1><p>This is running in port 82!</p>
root@VM1:~# echo '<h1> Welcome to my world Rishi</h1><p>This is running in port 82!</p>' | sudo tee /var/www/html/index.html
```

4. Deploy a sample index.html file on Apache.



5. Install Apache tomcat on port number 8082

```
root@VM1:~# sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-14 18:41:28 UTC; 12s ago
 Invocation: 9d551a5711c6471aa11d1d51c6c33c89
       Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
    Process: 11803 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 11809 (java)
      Tasks: 29 (limit: 1860)
     Memory: 167.7M (peak: 167.7M)
        CPU: 8.850s
     CGroup: /system.slice/tomcat10.service
             └─11809 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/logging.properties -Djava.uti

Oct 14 18:41:31 VM1 tomcat10[11809]: OpenSSL successfully initialized [OpenSSL 3.5.5 27 Jan 2026]
Oct 14 18:41:33 VM1 tomcat10[11809]: Initializing ProtocolHandler ["http-nio-8082"]
Oct 14 18:41:33 VM1 tomcat10[11809]: Server initialization in [3892] milliseconds
Oct 14 18:41:33 VM1 tomcat10[11809]: Starting service [Catalina]
Oct 14 18:41:33 VM1 tomcat10[11809]: Starting Servlet engine: [Apache Tomcat/10.1.55 (Ubuntu)]
Oct 14 18:41:34 VM1 tomcat10[11809]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
Oct 14 18:41:37 VM1 tomcat10[11809]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a
Oct 14 18:41:37 VM1 tomcat10[11809]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has finished in [3,786] ms
Oct 14 18:41:37 VM1 tomcat10[11809]: Starting ProtocolHandler ["http-nio-8082"]
Oct 14 18:41:37 VM1 tomcat10[11809]: Server startup in [4303] milliseconds

root@VM1:~# sudo ss -ltnp | grep :8082
LISTEN 0      100                *:8082             *:*    users:(("java",pid=11809,fd=37))
```



It works !

If you're seeing this page via a web browser, it means you've setup Tomcat successfully. Congratulations!

This is the default Tomcat home page. It can be found on the local filesystem at: `/var/lib/tomcat10/webapps/ROOT/index.html`

Tomcat veterans might be pleased to learn that this system instance of Tomcat is installed with `CATALINA_HOME` in `/usr/share/tomcat10` and `CATALINA_BASE` in `/var/lib/tomcat10`, following the rules from `/usr/share/doc/tomcat10-common/RUNNING.txt.gz`.

You might consider installing the following packages, if you haven't already done so:

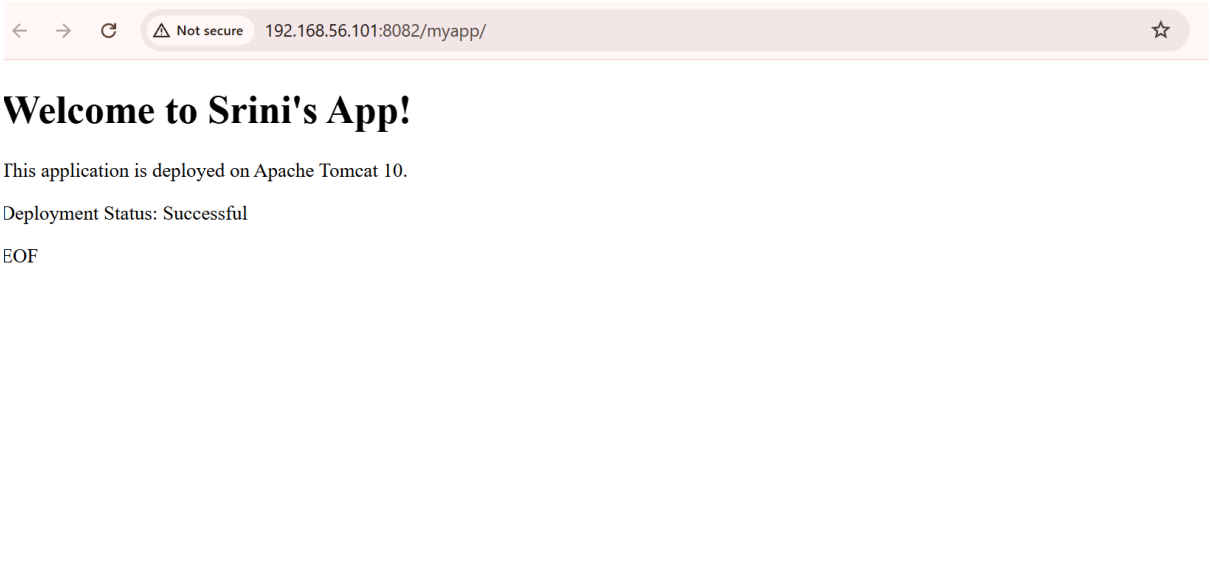
tomcat10-docs: This package installs a web application that allows to browse the Tomcat 10 documentation locally. Once installed, you can access it by clicking here.

tomcat10-examples: This package installs a web application that allows to access the Tomcat 10 Servlet and JSP examples. Once installed, you can access it by clicking here.

tomcat10-admin: This package installs two web applications that can help managing this Tomcat instance. Once installed, you can access the manager webapp and the host-manager webapp.

NOTE: For security reasons, using the manager webapp is restricted to users with role "manager-gui". The host-manager webapp is restricted to users with role "admin-gui". Users are defined in `/etc/tomcat10/tomcat-users.xml`.

6. Deploy a sample app on webapps

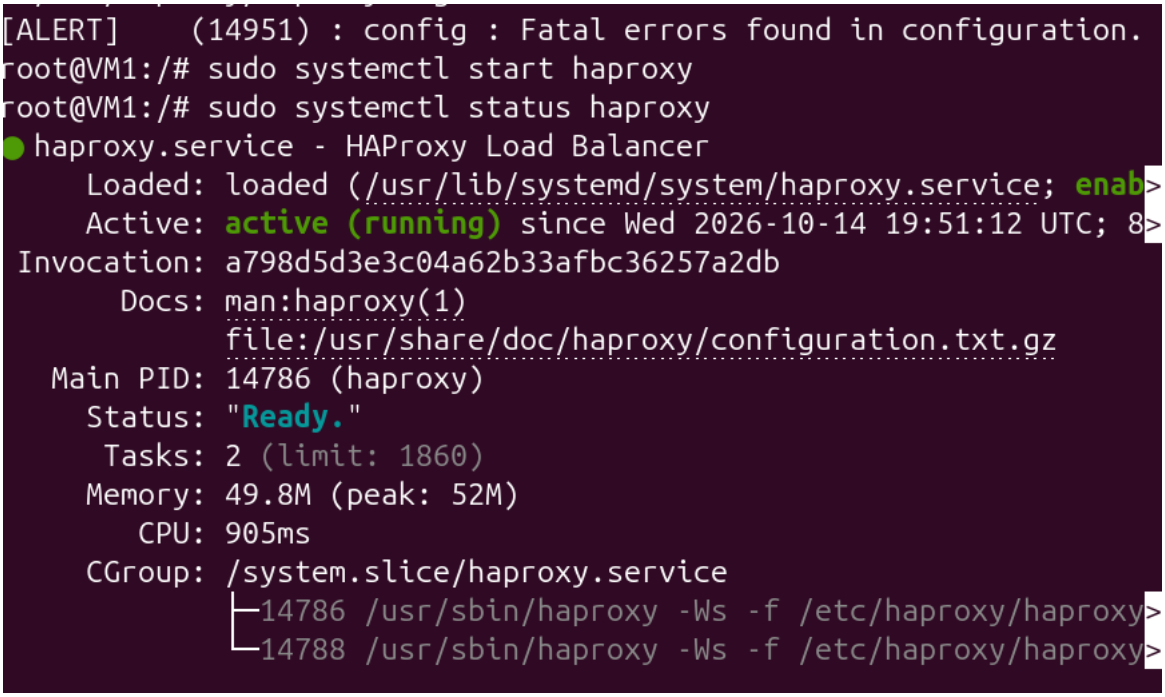


7. Create a tomcat.service file for tomcat.

```
oot@VM1:/# ls -ld /usr/share/tomcat10 /var/lib/tomcat10
rwxr-xr-x 5 root root 4096 Oct 10 11:37 /usr/share/tomcat10
rwxr-xr-x 5 root root 4096 Oct 14 19:15 /var/lib/tomcat10
oot@VM1:/# nano /etc/systemd/system/tomcat.service
oot@VM1:/# systemctl daemon-reload
oot@VM1:/# systemctl enable tomcat.service
reated symlink '/etc/systemd/system/multi-user.target.wants/tomcat.service' → '/etc/systemd/system/tomcat.service'.
oot@VM1:/# systemctl start tomcat.service
ob for tomcat.service failed because the control process exited with error code.
ee "systemctl status tomcat.service" and "journalctl -xeu tomcat.service" for details.
oot@VM1:/# systemctl status tomcat.service
 tomcat.service - Apache Tomcat10
     Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; >
     Active: failed (Result: exit-code) since Wed 2026-10-14 19:26>
Invocation: 41c5fadd5a0740408759ad4d95e0d430
    Process: 13536 ExecStart=/usr/bin/systemctl start tomcat10 (co>
   Mem peak: 1.8M
        CPU: 23ms

ct 14 19:26:30 VM1 systemd[1]: Starting tomcat.service - Apache T>
ct 14 19:26:30 VM1 systemctl[13536]: Failed to start tomcat10.ser>
ct 14 19:26:30 VM1 systemctl[13536]: See system logs and 'systemc>
ct 14 19:26:30 VM1 systemd[1]: tomcat.service: Control process ex>
ct 14 19:26:30 VM1 systemd[1]: tomcat.service: Failed with result>
ct 14 19:26:30 VM1 systemd[1]: Failed to start tomcat.service - A>
ines 1-14/14 (END)
```

8. Configure HA Proxy server


[ALERT] (14951) : config : Fatal errors found in configuration.
root@VM1:/# sudo systemctl start haproxy
root@VM1:/# sudo systemctl status haproxy
haproxy.service - HAProxy Load Balancer
Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enab>
Active: active (running) since Wed 2026-10-14 19:51:12 UTC; 8>
Invocation: a798d5d3e3c04a62b33afbc36257a2db
Docs: man:haproxy(1)
file:/usr/share/doc/haproxy/configuration.txt.gz
Main PID: 14786 (haproxy)
Status: "Ready."
Tasks: 2 (limit: 1860)
Memory: 49.8M (peak: 52M)
CPU: 905ms
CGroup: /system.slice/haproxy.service
14786 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy>
14788 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy>